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GIGA R1 Camera Guide

Learn about the GIGA R1 WiFi's camera connector, and how to stream data through Web Serial.

Author · Karl Söderby · Last revision · 07/23/2024

The GIGA R1 has a dedicated camera connector that allows certain camera modules to mount directly on the board. This makes it possible to add machine vision to your GIGA R1 board without much effort at all.

In this guide, we will explore the following:

Where the camera connector is located.

What cameras are compatible?

What library to use?

How to set up a camera stream to a browser using Web Serial.

Hardware & Software Needed

To follow and use the examples provided in this guide, you will need an [Arduino GIGA R1 WiFi](#)

You will also need the following software:

[Arduino IDE](#) (any version)

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Web Serial Web

Application (for displaying camera feed).

Supported Cameras

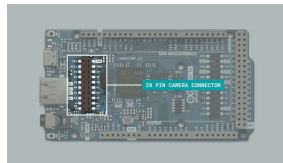
The GIGA R1 currently supports the following cameras, via the [Camera](#) library that is bundled with the [Arduino Mbed Core](#):

OV7670 and **OV7675**

GC2145

Himax HM01B0

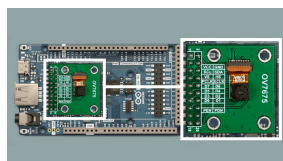
Camera Connector



Camera Connector on
GIGA R1

The 20 pin camera connector onboard the GIGA R1 is designed to be directly compatible with some breakout boards from ArduCam.

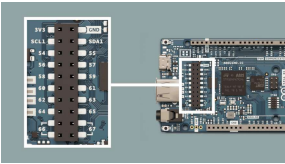
This allows you to simply connect the camera module directly to the board, without making any additional circuit.



Some of the 20 pin connector breakout boards from ArduCam can be found [here](#).

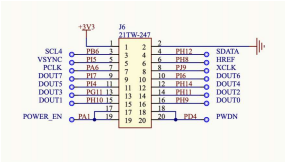
The complete pin map can be found below:

Left	Right
3V3	GND
SCL1	SDA1
54	55
56	57
58	59
60	61
62	63
64	65
66	67
66	67



Camera connector, zoomed in.

You can also view the schematic for this connector in more detail just below. This is useful to understand exactly which pins on the STM32H747XI microcontroller is used.



Schematic for Camera Connector (J6).

Raw Bytes Over Serial

This example allows you to stream the sensor data from your camera to a web interface, using serial over USB. This will allow you to see the image directly in your browser.

Step 1: Arduino

Upload the following sketch to your board.

This sketch is also available in the Arduino IDE via **Examples > Camera > CameraCaptureWebSerial**.

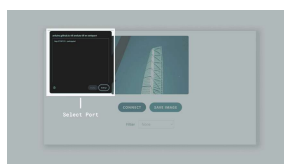


```
1  /*
2  * This example show
3  *
4  * There is a compan
5  * It can be found i
6  * The on-board LED
7  *
8  * Instructions:
9  * 1. Make sure the
10 * 2. Upload this sk
11 * 3. Open the web a
12 * in the "WebSerial
13 *
14 * Initial author: S
15 */
16
17 #include "camera.h"
18
19 #ifndef ARDUINO_NICLA
20 #include "gc2145.h"
21 GC2145 galaxyCore;
22 Camera cam(galaxyC
23 #define IMAGE_MODE
24 #elif defined(ARDUIN
25 // uncomment the c
26 #include "hm0360.h"
27 HM0360 himax;
28 // #include "himax
```

Step 2: Web Serial

Open the [Web Serial Interface](#) which allows you to view the camera feed. Make sure to close the Serial Monitor in the Arduino IDE beforehand, otherwise it will not work as the serial port would be occupied.

Press on **Connect** and select the correct port.



Select Port

There is also a variety of video filters that can be applied.



In this article, we learned a bit more about the camera connector on board the GIGA R1 board, how it is connected to the STM32H747XI microcontroller, and a simple example of how to connect ArduCam camera modules through web serial.

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